



Green Heaven

— PLANT FACTORY —



Spices & plants grown using organic methods
in controlled environment with high priority
to safety & hygiene

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DRY RACK
AEROPONIC



CONTROLLED
ENVIRONMENT
AGRICULTURE



PREMIUM
SPICES &
PLANTS



SUSTAINABLE
FUTURE



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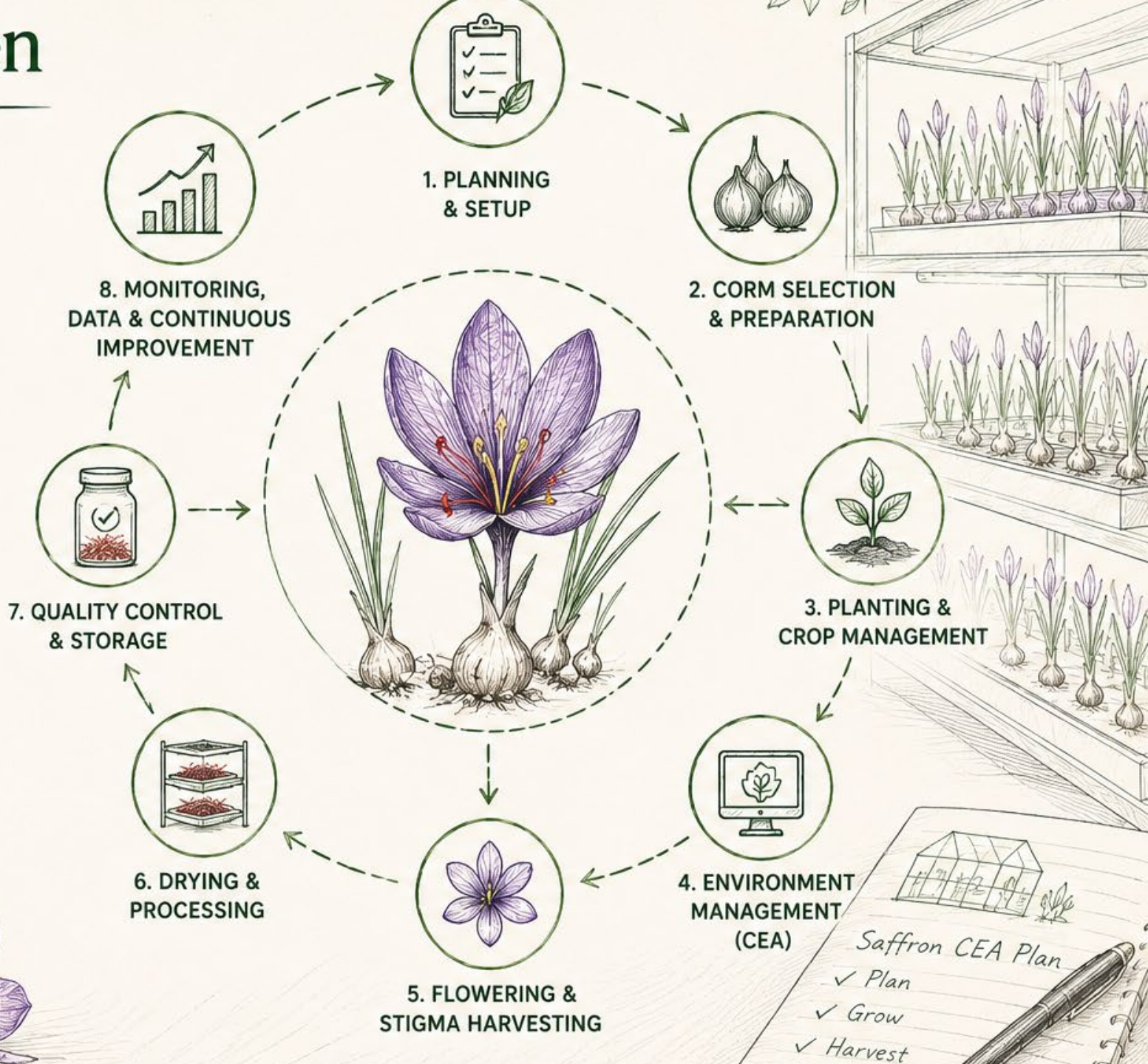
END TO END
**OPERATIONAL
PLAY BOOK**
FOR
SAFFRON
USING CEA



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Green Heaven Plant Factory

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Types of Cultivation Methods

Category	Attribute	Traditional Agriculture	Controlled Environment Agriculture	
Weather Source & Control	Light / Solar Energy	Natural Light 	Hybrid (Natural + LED)  	Artificial Spectrum (LED) 
	Temperature	No Control, natural variations 	Partial Control 	Complete Precision Automation 
	Humidity	No Control, natural variations 	Partial Control 	Complete Precision Automation 
	Carbon Dioxide (CO ₂)	No Control, natural variations 	Partial Control 	Complete Precision Automation 
Root Growth & Space	Growth Media	Soil 	Hydroponics (Preferred)    Soilless (Preferred) Soil (Rare)	Hybrid (Aeroponics + Soilless)  
	Level	Single Level 	Single Level (Preferred)  Vertical Stacking (Rare)	Vertical Stacking (Preferred)  Single Level (Rare)
	Space	Open Field 	Semi-transparent Glass / Plastic enclosure 	Non-transparent enclosure (Building / Container) 
Output	Extreme Weather Risk	High Risk	Medium Risk	No Risk
	Harvest Consistency	Low - Medium	Medium - High	High – Very High
	Land Use Efficiency	Low	Medium	Very High (due to vertical stacking)
			Topic of Focus	

Definition of Different Methods

	 Soil Media	 Soilless Media	 Aeroponics	 Hydroponics
Definition	Growing plants in soil , where the soil provides support, water, air, and nutrients to the roots.	Growing plants without soil , using an inert growing medium (such as coco peat, perlite, rockwool, or vermiculite) to support roots while nutrients are supplied through irrigation.	Aeroponics is a method of growing plants without soil or standing water , where roots are suspended in the air and periodically nourished by a highly atomized, oxygen-rich nutrient mist via humidifier	Growing plants without soil, with roots placed in a nutrient-rich water solution that supplies all required nutrients.
Ideal for	When good-quality land and soil are available, costs need to be kept low, and crops are grown in open fields or traditional greenhouses. Suitable for most field crops, fruit trees, and large-scale farming.	When growers want the benefits of controlled cultivation without the complexity of hydroponics. Well-suited for greenhouse vegetables such as tomatoes, cucumbers, peppers, flowers, and nursery plants using coco peat or similar media.	When maximum growth rates, root oxygenation, and water efficiency are desired. Commonly used for high-value crops, seed potato production, plant propagation, and advanced controlled-environment agriculture.	When water efficiency, high yields, and precise nutrient control are important. Ideal for leafy greens, herbs, strawberries, and greenhouse production in areas with poor soil quality.

Topic of Focus

Saffron Corm Key Characteristics & Make up

A **saffron corm** is the short, solid, fleshy underground stem of the *Crocus sativus* plant, which acts as its primary reproductive and energy-storage organ. Though it resembles a bulb, it is technically a corm—the foundational base from which the precious saffron spice grows

Key Characteristics

Structure: Saffron corms are typically egg- or disc-shaped, protected by a fibrous outer skin (tunic) that helps retain moisture during dormancy. [1, 2]

Function: They store the vital nutrients and carbohydrates a plant needs to survive through its summer dormancy and bloom during the fall months. [1, 2, 3, 4]

Propagation: The plant is sterile and cannot reproduce via seeds. Instead, the "mother" corm generates smaller "daughter" corms, which are harvested, separated, and replanted to grow new generations. [1, 2, 3]

Spice Yield: Each planted corm produces about 1 to 3 vivid violet flowers, and each flower yields only three red stigmas (the threads that make up the spice)



Buds: Points from where new shoots emerge

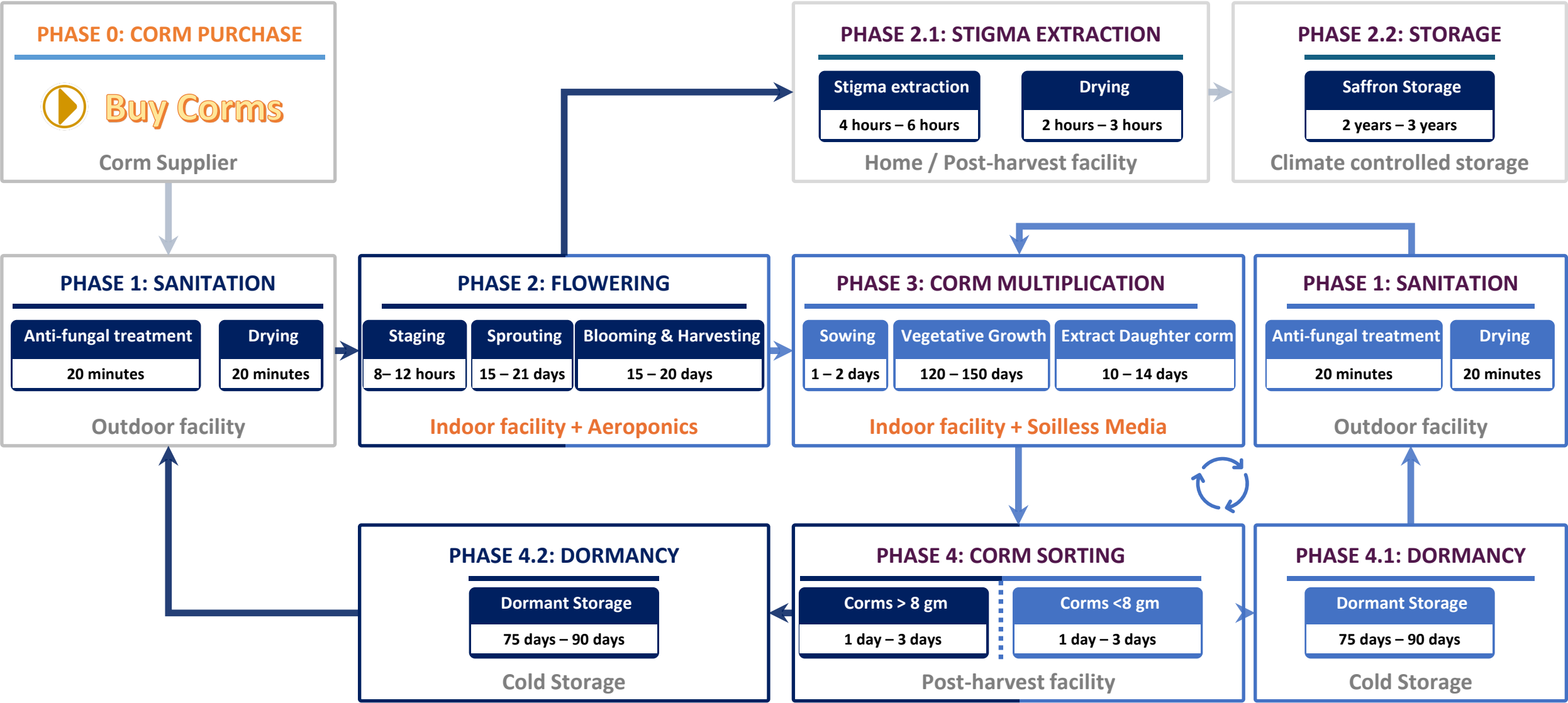
Nodes & Internodes: Compressed region that produce new corms & laves

Tunics: Dry, brown outer layers that protect the corm

Axillary buds: Located on sides, can develop into daughter corms

Roots: Fibrous roots emerge from basal plat to absorb water

Saffron Hybrid Farming Lifecycle



Phase-wise Functional & Technical Requirements

PHASE 0: CORM PURCHASE

PHASE DESCRIPTION:

Beginners will need to purchase Saffron Corms for first cycle. Post that, multiplication stage continues to provide for future corm requirements

CORM SIZE CONSIDERATION

- Purchase Jumbo / Premium size corms for first time staging
- Standard commercial is good if Jumbo is not affordable
- Do not buy under-sized corms / cormlets

Corm Size Group	Circumference (cm)	Weight (Grams)	Flowering Certainty	Expected Flower Count
Jumbo / Premium	11+ cm	20g to 25+ g	High (95%+)	2 to 4
Standard Commercial	9cm to 11 cm	10g to 19g	Medium to High	1 to 2
Under-Sized (Vegetative)	7cm to 8 cm	5g to 9g	Very low to None	0 to 1
Cormlets / Minis	Less than 7 cm	<5g	None	NA

CORM QUANTITY CONSIDERATION

- If starting new, recommend purchasing 500 – 1000 corms
- Multiplication over time will provide additional corms
- Table provides estimated quantity based on area available

Floor Area	Usable rack space	Total tray space (5 tiers)	Total corm count needed (full capacity)
100 square feet	55 square feet	275 square feet	~6,875
200 square feet	110 square feet	550 square feet	~13,750
300 square feet	165 square feet	825 square feet	~20,625

PURCHASE CONSIDERATIONS

- Buy corms that are not aged beyond 90 days after dormancy
- Suppliers tend to promise one size while paying, but, ship mixed sizes while shipping
- Buy a smaller quantity (50 – 100) to sample test supplier before committing to 500 – 1000 quantity
- Best to visit the seller in person & purchase at the facility than courier option
- Purchase organic corms if intent is to grow Saffron using organic methods
- You may need to validate organic certification before purchasing corms from supplier

Phase-wise Functional & Technical Requirements

PHASE 1: SANITATION

PHASE DESCRIPTION: Whether buying corms from supplier or using after dormancy phase, corms tend to carry fungal infections. It is a mandatory step to sanitize the corms before staging for flowering

SANITATION OPTIONS

- For conventional method: Use chemicals listed
- For organic method: Use pure, cold pressed neem oil with organic liquid soap
- Solution can be used to sanitize 3 batches of corms

Sanitation Type	Water	Carbendazim (50 WP)	Mancozeb (75 WP)	Cold pressed Pure Neem Oil	Organic liquid soap	Corm batch size	Total Corms for 3 batches
Conventional	1 Litre	1 gram	3 grams	NA	NA	~10 to 15	~30 to 45
Organic	1 Litre	NA	NA	5 ml to 7.5 ml	2 ml to 3 ml	~10 to 15 Corms	~30 to 45

MIXING PROCESS

CONVENTIONAL METHOD:

- Take desired quantity of water in a container
- Add chemicals based on the proportion provided above
- Stir the mixture for 5 minutes

ORGANIC METHOD:

- Measure neem oil & liquid soap & pour into a small container first
- Mix rigorously till oil turns cloudy & forms uniform soap paste
- Pour mixture into warm water and stir
- If you see oil drops floating on top, add a tiny bit more soap to finish breaking down the oil

SANITATION & DRYING

- Add corms to the prepared solution
- Let corms soak for 20 minutes
- Remove the corms & dry in shade for 20 minutes
- This only ensures corms are not dripping the solution, further drying is needed before staging on trays

Phase-wise Functional & Technical Requirements

PHASE 2: FLOWERING (PART1)

PHASE DESCRIPTION: Before staging the sanitized corms for blooming phase, an indoor climate controlled facility is needed, Part 1 focuses on providing the infrastructure details needed for this phase

INDOOR FACILITY INFRASTRUCTURE CONSIDERATIONS:

Infrastructure Element	Functional Specifications	Description	Where can I find the product?
Space	Fully insulated, enclosed space	Cleanroom environment. Use wash-safe PUF panels. Prevents wild mold spores	Local Fabrication / PUF Panel Supplier
Adjustable Racks	Multi-tier powder-coated steel racks	18 to 24 inches vertical clearance per tier	Local Fabrication / PUF Panel Supplier
HVAC (Heating, Ventilation, Air Conditioning) System	Industrial inverter AC unit. Must maintain cooling down to 8°C. High vertical airflow.	The cooling unit must be able to maintain its low target continuously without short-cycling	For mechanical operations: Pioneer Split System For integrated automation: AC Infinity
Air Circulation	Continuous vertical air velocity of 0.2 to 0.5 m/s inside the foliage. Rack level fans: One fan every 4 – 6 feet Room level fans: 100 sq ft room: 1 Exhaust Fan (6-inch dia, 300–400 CFM) 200 sq ft room: 1 Exhaust Fan (8-inch dia, 700–800 CFM) 300 sq ft room: 2 Exhaust Fans (6 or 8-inch) mounted at opposite corners of the ceiling to prevent dead zones.	Breaks boundary layers and prevent stagnant microclimates. Digital Anemometer (Air Velocity Meter) is needed to measure air velocity	Continuous Air Circulation (Rack Level): Infinity Duct/Exhaust Air Exchange (Room Level): AC Infinity , AC Infinity Intelligent
LED System	Color spectrum: Full-Spectrum White with enhanced Red (600-660nm) and Blue (450-470nm) peaks Light intensity: Maintain a Photosynthetic Photon Flux Density (PPFD) of 100 to 150 $\mu\text{mol}/\text{m}^2/\text{s}$ at the tray surface	Standard household lights lack the intensity needed to trigger flower pathways. Look for specialized commercial-grade bar fixtures	Small Space / Pilot: Samsung For large operations & Automation: MARS HYDRO MARS SMART
Humidifier	Industrial ultrasonic mist humidifiers. Connect directly to digital humidistats.	Saffron rooms require thick, cool fog. Avoid heat-generating steam models	Small Space / Pilot: FOGG PRO For large operations: 25 Kg/hr, high capacity humidifiers
Water quality for humidifier	Requires Reverse Osmosis (RO) or Demineralized water with a Total Dissolved Solids (TDS)	Maintain rating under 10 PPM to prevent mineral dust buildup on corms and equipment	Backup power systems
Power Back up	Prevent power failure to avoid high temperature, which results in corms aborting flowering cycle	Install independent, cloud-connected backup temperature/humidity sensors to send immediate phone alerts if the primary environment system fails	
Automation Sensors	Multi-point data collection array	Install independent sensor probes on the top, middle, and bottom rack tiers to calculate a true room average.	AC Infinity Smart UIS Sensor Probes
Manual Back up	A high-precision, calibration-certified thermo-hygrometer. It should be mounted on the wall furthest from your HVAC output at eye level	Battery operated wall hygrometer as back up in case of automated sensor malfunction	HTC2 Taylor

Phase-wise Functional & Technical Requirements

PHASE 2: FLOWERING (STAGING)

PHASE DESCRIPTION: Staging for blooming essentially means to arrange sanitized & dried corms in trays organized on a rack system

TECHNICAL & FUNCTIONAL CONSIDERATIONS:

- Material option is same whether going for conventional or organic
- Perforated trays ensure 360 degree air and humidity movement

Parameter	Requirement	Technical Specification & Justification
Material & Structure	Food-Grade, Non-Porous HDPE or PP Plastic with a Perforated Mesh Bottom.	Prevents water retention. Solid plastic or wooden trays trap dampness and harbor latent fungal spores
Tray Dimensions	Option A (Standard Euro-Crate): 600 mm × 400 mm (approx. 24 × 16 inches) Option B (Standard USA Flat): 20 × 10 inches. Ideal for tighter rack shelves	Fits standard vertical farming racks perfectly. Sized to hold 140–160 corms per tray
Tray Depth	The ideal mechanical tray depth is 2.5 to 3 inches (6.5 cm to 7.5 cm)	If a tray is too shallow (under 1.5 inches), the corms will tumble, roll, and crowd each other when racks are moved

TRAY SANITIZATION & CORM STAGING:

- **Conventional method:** Trays must be rinsed with water after treatment & fully dried before staging corms
- **Organic method:** No rinsing is needed, but, trays must be dried before staging corms
- **Staging corms in trays:** Ensure spacing of 1 cm to 1.5 cm of gap on all sides
- Ensure corms are fully dry as well before staging on trays

Factor	Conventional Method	Organic Method
Primary Sanitizer	Sodium Hypochlorite (Chlorine Bleach)	Peracetic Acid (PAA) OR Hydrogen Peroxide (H ₂ O ₂)
Mixing Ratio	10% Bleach to 90% Tap Water	0.5% to 1% PAA solution OR 3% pure H ₂ O ₂

OPERATIONAL CONSIDERATIONS:

- **Size Segregation:** Stage similar sized corms in same trays (All jumbo in one tray vs all Standard in different tray) since bigger corms grow faster compared to smaller ones
- **Labelling:** Labelling trays with important information goes a long way in effective operational controls & metrics tracking over time. Provided is an example of labelling, though this is not exhaustive & can be modified based on specific needs

BATCH / LOT ID:	B2-YYYY-CITY-04 (Track supplier)	
STAGING DATE:	DD-MM-YYYY (Calculate flowering cycle)	
CORM GRADE:	JUMBO (22g – 25g) (Calculate yield by grade)	
CORM COUNT:	152 (Calculate yield by count)	
METHOD:	ORGANIC (Brand positioning)	

Phase-wise Functional & Technical Requirements

PHASE 2: FLOWERING (SPROUTING)

PHASE DESCRIPTION: Once staging is done, it is time to create ideal indoor conditions for corms to wake up & start sprouting. At end of this cycle, the room looks like a grid of white spikes with no visible flowers yet

INDOOR CONDITIONS CONSIDERATION:

- For full duration of 15 – 21 days, keep LED lights completely OFF
- No nutrients needed, corms use stored nutrients for sprouting
- Keep room level exhaust at 2 – 3 room air exchange per hour always

Schedule	Temperature	Relative Humidity (RH)	Rack level fans	Duration
Day Time	16°C – 18°C	75%	0.2 m/s	10 hours
Night Time	13°C – 15°C	80%	0.4 m/s	14 hours

DAILY OPERATIONAL CONSIDERATIONS:

- Inspect tray bottoms for early root popping:** If root popping is spotted, change **NIGHT TIME** settings (No change to day time settings):
 - Humidity:** From 80% to 85%
 - Rack level fan speed:** From 0.4 m/s to 0.5 m/s.
- Cross-verify wall hygrometers against sensor logs:** If mis-alignment is spotted, technical inspection of sensors may be needed
- Minimize entry into the room to preserve darkness & temperature stability
- Inspection for & removal of rotting corms:**
 - The Smell Test:** If the room smells like rotten onions instead of clean earth, an outbreak is happening.
 - The Eye Test:** Look for any corms developing a white, pink, or blue fuzzy mold on their outer burlap skin, or corms leaking a yellowish liquid
 - The Touch Test:** Gently pinch **suspect corms** only. A healthy corm feels rock-hard like a potato. A rotting corm feels soft, squishy, and hollow.

Phase-wise Functional & Technical Requirements

PHASE 2: FLOWERING (BLOOMING & HARVESTING)

PHASE DESCRIPTION: Focuses on maximizing synchronous flower emergence and protecting stigma quality.

INDOOR CONDITIONS FOR BLOOMING:

- Keep the lights off during night
- No nutrients needed, corms use stored nutrients for blooming
- Keep room level exhaust at 2 – 3 room air exchange per hour always

Schedule	Temperature	Relative Humidity (RH)	Rack level fans	Lighting Intensity	Duration
Day Time	11°C – 13°C	75% - 80%	0.2 m/s	100 to 150 µmol/m²/s	10 hours
Night Time	8°C – 10°C	80% - 85%	0.5 m/s	LIGHTS OFF	14 hours

HARVESTING CONSIDERATIONS:

- **The Morning Pluck:**
 - Harvest all flowers in the early morning at the "Closed Bud" or semi-open stage
 - Harvest all flowers before lights cycle on or within the first hour of lights cycle
 - Snap stems cleanly at the base.
- **The 4-Hour Clock:** Stigma extraction must be completed within 4 hours. Transfer flowers to food-grade containers and move to post-harvest instantly.
- **Sterilization Cycle:** Execute a daily audit; instantly cull soft, moldy, or blind corms into sealed bags to preserve room sterility
 - **The Smell Test:** If flowers smells like rotten onions instead of sweet honey-like smell, an outbreak is happening.
 - **The Eye Test:** Look past the bright purple petals down to the tray mesh floor. You are hunting for fuzzy grey mold (Botrytis) or white mold (Fusarium) attacking the base of the flowering stalks or leaking a yellowish fluid
 - **The Touch Test (The Pinch):** If a corm has pushed out a leaf shoot but failed to produce a flower (a "blind" corm), or if the flower looks dried up and withered before it even opened, a worker must gently pinch the corm. If it feels soft and squishy, it must be culled immediately.

Transition from Aeroponics to Soilless Media

PHASE DESCRIPTION: Once flowers are plucked, 2 key activities need to take place in parallel:

- 1. **Within 24 – 48 hours:** Re-plant corms to soilless media immediately
- 2. **Within 4 hours:** Extract Stigma from plucked flowers (Continued in slide xx)

In this slide, we will cover available & recommended options for soil-less media for re-planting Corms.

SUBSTRATE (SOILLESS MEDIA) MIX OPTIONS:

Preference	Substrate Mix Style	Organic Status	Technical Performance Matrix	Saffron Risk / Benefit Evaluation
HIGHLY RECOMMENDED	70% Cocopeat 30% Perlite	Fully Organic <i>(With Certified Cocopeat)</i>	Excellent. Premium water holding capacity matched with instant drainage.	LOW RISK: The perlite creates open oxygen pockets that prevent the cocopeat from compacting over the 150-day cycle.
ALTERNATIVE OPTION	70% Cocopeat 30% Vermiculite	Fully Organic <i>(With Certified Cocopeat)</i>	Very High-Water Retention. Holds onto liquid nutrients exceptionally well.	MEDIUM RISK: Vermiculite retains significantly more water than perlite and compresses over time. If your sprinklers over-irrigate by even a few minutes, this mix turns muddy and causes root rot.
NOT RECOMENDED	70% Sandy Loam Soil 30% Perlite	Fully Organic <i>(Must be Oven-Sterilized or Steam-Pasteurized)</i>	High moisture and nutrient retention, but lacks the fluffy consistency of cocopeat and remains highly labor-intensive	HIGH RISK: Raw outdoor soil (especially clay-heavy Red Soil) must never be used. It causes severe compaction under indoor sprinklers, blocking root oxygen and harboring latent soil-borne fungal pathogens (Fusarium) that can decimate your racks
NOT RECOMMENDED	100% Pure Cocopeat	Fully Organic <i>(With Certified Cocopeat)</i>	High moisture retention but zero structural drainage.	HIGH RISK: Without perlite or vermiculite to break it up, pure cocopeat packs down tightly under sprinkler fire, suffocating the roots from oxygen.

Phase-wise Functional & Technical Requirements

PHASE 3: CORM MULTIPLICATION (PART 1)

PHASE DESCRIPTION: Before re-planting corms into soilless media for multiplication phase, an indoor climate controlled facility is needed, Part 1 focuses on providing the infrastructure details needed for this phase

INDOOR FACILITY INFRASTRUCTURE CONSIDERATIONS:

Infrastructure Element	Functional Specifications	Description	Where can I find the product?
Space	Fully insulated, enclosed space	Use wash-safe PUF panels. Walls must resist constant humidity and prevent moisture trap behind racks	Local Fabrication / PUF Panel Supplier
Adjustable Racks	Multi-tier powder-coated steel racks equipped with slanted wastewater tracks	18 to 24 inches vertical clearance per tier. Install rigid plastic drain trays or vinyl collection channels directly below each rack tier to route sprinkler runoff away safely	Local Fabrication / Greenhouse Supplier
Automated Sprinkler System	Overhead Micro-Misting Sprinklers or specialized Drip Irrigation Stakes	Connected to a heavy-duty digital solenoid valve timer. Maintain TDS rating under 150 PPM for sprinklers to ensure clean organic NPK blending	Micro Drip Irrigation Kit
HVAC (Heating, Ventilation, Air Conditioning) System	Industrial inverter split AC unit with heating module	Must maintain a warm, artificial spring cycle between 14°C and 22°C. System requires variable compressor modulation to handle heat spikes from long grow light hours	For mechanical operations: Pioneer Split System For integrated automation: AC Infinity
Air Circulation	Continuous vertical air velocity of 0.3 m/s inside the foliage. Rack level fans: One fan every 4 – 6 feet Room level fans: 100 sq ft room: 1 Exhaust Fan (6-inch dia, 300–400 CFM) 200 sq ft room: 1 Exhaust Fan (8-inch dia, 700–800 CFM) 300 sq ft room: 2 Exhaust Fans (6 or 8-inch) mounted at opposite corners of the ceiling to prevent dead zones.	Keeps the dense, grassy leaf rows moving horizontally & prevents dampness. Place fans horizontally across tray beds.	Continuous Air Circulation (Rack Level): AC Infinity , CLOUDRAY Clip Fan Duct/Exhaust Air Exchange (Room Level): AC Infinity , CLOUDNINE T6 / T8

Infrastructure Element	Functional Specifications	Description	Where can I find the product?
LED System	Color spectrum: Full-Spectrum White with heavy vegetative blue peaks Light intensity: Maintain a PPFD of 150 to 200 $\mu\text{mol}/\text{m}^2/\text{s}$ at the leaf tip	Requires higher light intensity than the flowering phase to drive photosynthesis	Small Space / Pilot: Samsung , SPIDER FARMER For large operations & Automation: MARS HYDRO , MARS SMART
Humidifier	Industrial ultrasonic mist humidifiers. Connect directly to digital humidistats.	Used to maintain ambient leaf turgor without over-saturating the potting medium	Small Space / Pilot: FOGG PRO , Saffron Humidifier For large operations: 25 Kg/hr, high capacity humidifiers
Water quality for humidifier	Requires Reverse Osmosis (RO) filtration	Maintain a TDS rating under 10 PPM for the humidifier	
Power Back up	Prevent power failure to avoid heat spikes	Install independent, cloud-connected backup temperature/humidity sensors to send phone alerts if primary environment system fails	Backup power systems
Automation Sensors	Multi-point data collection array	Install independent sensor probes on the top, middle, and bottom rack tiers to calculate a true room average.	AC Infinity Smart , UIS Sensor Probes

Phase-wise Functional & Technical Requirements

PHASE 3: CORM MULTIPLICATION (SOWING 1/2)

PHASE DESCRIPTION: After flowering, mother corms need to be replanted in cocopeat bed. Replanting has 3 critical process steps:

- A. SUBSTRATE MIX PREPARATION
- B. NUTRIENT SOLUTION PREPARATION
- C. SOWING CORMS

A. SUBSTRATE MIX PREPARATION

- Step 1:** Prepare acidic solution with pH level 5.5 – 6.2 (Use pH meter)
- Step 2:** Soak cocopeat overnight for 12 – 16 hours
- Step 3:** Squeeze moisture to 60% (Coir should ball up perfectly in your fist when squeezed hard and release only 2 to 3 drops of water)
- Step 4:** Blend with perlite at 70% to 30% ratio

Note: 5 kg cocopeat block requires 20 to 25 Liters of water to fully hydrate

ORGANIC		CONVENTIONAL	
COMPONENTS	QUANTITY	COMPONENTS	QUANTITY
Pure RO Water	1 Litre	Pure RO Water	1 Litre
Citric Acid (Powder)	1 Pinch	Phosphoric Acid (85% Liquid)	Few Drops
OMRI-Listed Liquid Organic Calcium	4 – 6 ml	Calcium Nitrate	1.5 gm

B. NUTRIENT SOLUTION PREPARATION

- Step 1:** Prepare nutrient solution in a tank
- Step 2:** Connect tank to sprinkler system
- Step 3:** Measure Electrical Conductivity (EC):
- First 2 weeks: 0.8 mS/cm
 - Post 2 weeks: 1.2 – 1.4 mS/cm

Note: 100 sq ft room (275 sq ft tray space) = ~41 Liters of nutrient solution per day

NUTRIENTS	ORGANIC	CONVENTIONAL	BIOLOGY FUNCTION
Water Base	100 Litres Pure RO Water	100 Litres Pure RO Water	Hydration & dilution of nutrients
Nitrogen (N)	40 Grams Water-Soluble Soy Protein Hydrolysates or Amino Powders.	30 Grams Synthetic Urea OR Calcium Nitrate.	Drives early spring foliage elongation to power photosynthesis.
Phosphorus (P)	20 Grams Liquid Organic Bone Meal extract or Micronized Rock Phosphate (<50 microns).	25 Grams Mono-Potassium Phosphate (MKP).	Stimulates dense root anchoring and cellular energy division.
Potassium (K)	80 Grams highly soluble, OMRI-listed Mined Potassium Sulphate (K ₂ SO ₄).	90 Grams Commercial Potassium Nitrate (KNO ₃) or Muriate of Potash.	The Core Driver. Packs carbohydrate mass into developing daughter corms.
Micronutrient Pack	100 ml Liquid Cold-Processed Seaweed/Kelp extracts + 15 Grams Azomite powder	10 Grams Synthetic EDTA Chelated Iron, Zinc, Manganese, and Boron salts	Synthesizes crucial plant enzymes and protects leaves from chlorosis (yellowing).

Phase-wise Functional & Technical Requirements

PHASE 3: CORM MULTIPLICATION (SOWING 2/2)

PHASE DESCRIPTION: Continuation from prior slide to cover Sowing corms for multiplication

TRAY SPECIFICATIONS:

- Material option is same for conventional OR organic

Parameter	Requirement	Technical Specification & Justification
Material & Structure	Food-Grade, Non-Porous High-Density Polyethylene (HDPE) or Polypropylene (PP) heavy-duty plastic with a Solid Bottom or Precision Drainage Slits	Unlike mesh flowering trays, these must hold loose substrate. Solid-bottom containers require a drill pattern or integrated drainage slots
Tray Dimensions	Option A (Standard Euro-Trough Crate): 600 mm × 400 mm (approx. 24 × 16 inches)	Fits standard vertical farming rack systems perfectly and is easily handled by a single worker
	Option B (Long-Profile Linear Planter): 1000 mm × 300 mm (approx. 39 × 12 inches)	Allows for continuous linear row planting, maximizing the sprinkler footprint on long rack tiers
Tray Depth	8 to 10 inches (20 cm to 25 cm) depth to support a minimum substrate depth of 6 to 8 inches (15 cm to 20 cm)	Critical for multiplication.

REPLANTING CORMS (SOWING):

- **Tray Baseline:** Shovel a baseline of 3 inches (7.5 cm) of the prepared 70/30 substrate mix
- **Corm Placement:** Place corms with sprout pointing upward
- **Planting Distance:** Ensure a gap of 5 – 7 cm on all sides of corms
- **Burying Corms:** Top up corms with additional 4 - 5 inches (10 to 12 cm) of the prepared 70/30 substrate mix
- **Initial Nutrient Supply:** Fire automated overhead sprinklers for 3 to 4 minutes using pure, pH-adjusted RO water (pH 5.5–6.2, No NPK).

SUBSTRATE MIX QUANTITY CONSIDERATIONS:

Depending on the tray dimensions, here is the exact substrate mix weight required for sowing corms:

- **Euro Crate (600 mm × 400 mm):** 11 kg to 14 kg per tray
- **Linear Planter (1000 mm × 300 mm):** 13.5 kg to 18 kg per tray

PHASE 3: MULTIPLICATION (VEGETATIVE GROWTH)

INDOOR CONDITIONS CONSIDERATION:

- ## IRRIGATION CONSIDERATIONS:

- | Schedule | Specification | pH Level | NPK Ratio | NPK Solution | EC Target |
|---------------|---------------------|-----------|-----------|---------------|---------------|
| First 2 Weeks | Mild Nutrient Water | 5.5 – 6.2 | 2:1:4 | Half Strength | 0.8 mS/cm |
| Post 2 Weeks | Rich Nutrient Water | 6.0 – 6.5 | 2:1:4 | Full Strength | 1.0–1.2 mS/cm |

Filter Flush Protocol: Manually flush the 50-micron inline irrigation disc filter once every week to prevent organic amino or rock phosphate sediments from clogging your overhead sprinkler nozzles.

Phase-wise Functional & Technical Requirements

PHASE 3: MULTIPLICATION (EXTRACT DAUGHTER CORMS)

PHASE DESCRIPTION: Post vegetative phase, it is time now to start extracting daughter corms.

THE DROUGHT TRIGGER:

- Follow this schedule for 10 – 14 days

Schedule	Temperature	Relative Humidity (RH)	Rack level fans	Lighting	Light Intensity	Sprinklers
24 Hours	22°C – 25°C	40% - 50%	0.5 m/s – 0.6 m/s	OFF	NA	OFF

DAUGHTER CORM EXTRACTION: Start extraction process around 11th day by overturning the tray contents on to a mesh table. Sanitize the daughter corms using below process:

De-Hulling & Core Trimming: Twist off loose outer husks & clip dead root strings. Scrape off dried plate remnants of the old mother corm from the base of the new daughter corms

The Dry Surface Brush: Use a soft-bristled brush to sweep any micro-particles of cocopeat dust inside skin folds.

The Protective Powder Dust (Optional Best Practice): Lightly dust the corms with an organic, fine-grade sulphur powder or a dry bio-fungicide powder containing *Trichoderma harzianum*.

The Open-Air Curing Check: Spread the trimmed, sorted corms in a single layer on clean plastic mesh trays for 24 hours under max fan circulation before closing the storage doors.

PHASE 4: CORM SORTING

CORMS > 8g: Move corms to aeroponics facility & start dormancy period

Schedule	Temperature	Relative Humidity (RH)	Room Level Exhaust	Lighting
24 Hours	25°C	65%	0.5 – 1.0 Air exchange / hour	OFF

CORMS < 8g: Keep corms in multiplication facility & start dormancy period

Schedule	Temperature	Relative Humidity (RH)	Room Level Exhaust	Lighting
24 Hours	25°C	40% - 50%	0.5 – 1.0 Air exchange / hour	OFF

Phase-wise Functional & Technical Requirements

PHASE 2.1: STIGMA EXTRACTION (1/2)

PHASE DESCRIPTION: After harvesting flowers, Stigma (Saffron) must be extracted within 4 - 6 hours

EXTRACTION SET-UP CONSIDERATIONS:

- **Workstation:** Cleanroom workstation with stainless steel tables, wipeable surfaces, & ~18°C temperature to prevent the harvested flowers from wilting.
- **Hygiene:** Must wear hairnets, face masks, & powder-free nitrile gloves to prevent standard skin oils and sweat from contaminating the gourmet spice
- **Collection Tools:** Food-grade, shallow stainless steel trays or BPA-free plastic bins lined with parchment paper.

STIGMA (SAFFRON) EXTRACTION CONSIDERATIONS:

- **4-Hour Time Limit:** Extraction must occur within 4 to 6 hours of plucking the flowers from the growth chamber
- **Physical Split:** Open the purple petals. Locate where the 3 red threads merge into a single yellow tube at the stem base.
- **Precision Pinch:** Pinch the three red filaments right above the point they turn yellow. Avoid pulling the yellow style out with it (dilutes the coloring strength)

STIGMA GRADING:

- **Grade 1 (Mongra/Sargol):** Exclusively deep crimson-red tips of the stigma. It has zero yellow style material attached.
- **Grade 2 (Lacha/Pushal):** Includes the 3 red filaments still connected to a short 2–3 mm piece of the yellow style.
- **Grade 3 (Guncha/Bunch):** The full red stigmas tied in bundles with the entire long yellow style intact.

STIGMA DRYING (DEHYDRATOR): RECOMMENDED

- **Hardware:** COSORI OR FATHOM
- **Drying:** Lay threads flat in a single layer on parchment-lined trays. Dry at 45°C to 50°C for 2 to 3 hours
- **The Target Check:** The threads should be brittle, dark red, and break with a clean snap when touched

STIGMA DRYING (SHADE): NOT RECOMMENDED

If forced to use this option: Spread threads out very thinly on silk mesh screens in a dedicated room held at 20°C with an aggressive 35% RH ceiling. Run room-level inline exhaust fans continuously at 5–6 air exchanges per hour to force fast evaporation and prevent stagnant rot. This slow process takes 6 – 8 days.

Phase-wise Functional & Technical Requirements

PHASE 2.1: STIGMA EXTRACTION (2/2)

PHASE DESCRIPTION: The purple flower petals (Floral Biomass) & yellow stamen (Pollen Pod) are valued by cosmetic industry, requires drying & storage

FLORAL BIOMASS & POLLEN PODS DRYING & STORAGE CONSIDERATIONS:

- Petals (Floral Biomass):** Saffron petals contain heavy moisture and volatile anthocyanins. Petals must be dried in dehydrator with 45°C to 50°C.
- Stamens (Pollen Pods):** Stamens are small and hold loose pollen. They must be dried on fine parchment paper liners or ultra-fine silk mesh screens.
- The Compaction Factor:** Dried purple petals take up massive volumetric space. Commercial facilities use heavy-duty Airtight Food-Grade Vacuum Sealers to compress the dried petals into dense brick blocks before storing them in opaque storage bins.
- Pollen Preservation:** Dried stamens are highly hygroscopic (they aggressively suck moisture out of the air). They must be locked away instantly with food-safe silica gel desiccant packets dropped directly inside the airtight glass or aluminum containers to preserve the natural yellow pigment properties.

PARAMETER	PURPLE PETALS (FLORAL BIOMASS)	YELLOW STAMEN (POLLEN POD)
Drying Method	Forced-Air Oven / Dehydrator	Forced-Air Oven / Dehydrator
Temperature Target	40°C to 50°C (Max 60°C)	45°C to 50°C
Drying Duration	4 to 6 Hours (Until crisp)	2 to 3 Hours (Dries faster than petals)
Target Moisture	8% to 10%	6% to 8%
Storage Container	Vacuum-sealed thick UV-opaque bags: MJ Commercial , Out of Air (OR) Food-grade airtight drums: VTC , Vintage	Airtight amber glass jars: Eisco , Borosilicate (OR) Dark aluminum storage tins: Healthcom
Storage Climate	Cool & Dry: 15°C to 18°C with RH < 45%.	Cool & Dry: 15°C to 18°C with RH < 40%.
Ideal Shelf Life	12 to 18 Months	12 to 18 Months

Phase-wise Functional & Technical Requirements

PHASE 2.2: SAFFRON STORAGE

PHASE DESCRIPTION: Once extracted & dried, saffron strands can have a shelf life of 2 – 3 years

SAFFRON HOME STORAGE CONSIDERATIONS:

- **<40% Relative Humidity:** To achieve this without industrial equipment, drop an OMRI-listed, food-grade 2g [silica gel packet](#) into each airtight container before locking to automatically absorb residual internal moisture.
- **The Ideal Location:** A cool, dark, interior pantry, closet, or kitchen cabinet that is situated away from exterior walls, windows, stove heat, and direct sunlight
- **Places to Avoid:** Near the cooking stove, oven, or dishwasher, as these appliances generate massive spikes of local heat and steam that will heat up the jars and ruin the spice volatiles.
- **The Blackout Box Best Practice:** Place your amber jars or aluminum tins inside a dark, thick plastic storage tote box or a wooden box inside that closet. This creates a "Double-Insulation Barrier" that completely shields the containers from any ambient light and buffers them against home temperature shifts

PARAMETER	OPERATIONAL TARGET METRIC	FUNCTIONAL JUSTIFICATION & MATERIAL CONTROL
Primary Enemies	Light, Air, & Moisture	Exposure to light destroys aroma; air exposure breaks down color.
Container Style	Airtight, opaque containers	Airtight amber glass jars: Eisco , Borosilicate (OR) Dark aluminum storage tins: Healthcom
Room Temperature	15°C to 18°C	Cool ambient levels prevent chemical volatile evaporation.
Relative Humidity	Strictly under 40% RH	Prevents the dry spice from absorbing ambient moisture.
Total Product Shelf-Life	2 to 3 Years	Saffron remains at premium grading if held under these clean vault settings.



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5 YEAR FINANCIAL MODEL FOR SAFFRON USING CEA

A PRACTICAL ROADMAP FOR BEGINNERS
TO START & SCALE WITH CONFIDENCE



BEGINNER
FRIENDLY



5 YEAR
PROJECTIONS



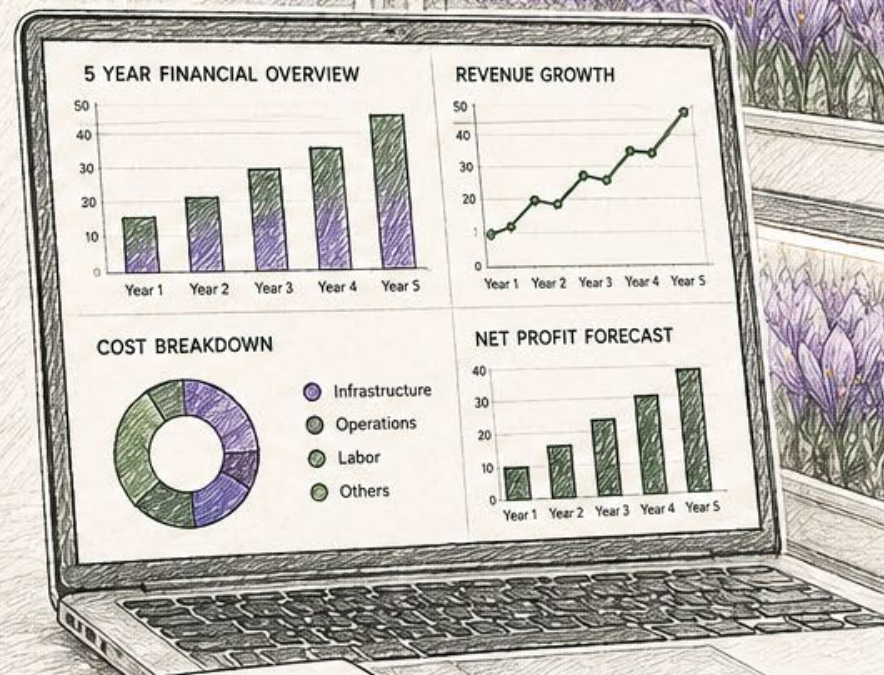
COST, REVENUE &
PROFIT ANALYSIS



RISK & RETURN
ASSESSMENT



SUSTAINABLE
GROWTH PATH



CEA SETUP
& INVESTMENT
PLANNING



OPERATING
EXPENSES &
MANAGEMENT



REVENUE
STREAMS &
PRICING



PROFITABILITY
& CASH FLOW
ANALYSIS



5 YEAR GROWTH

Ajith Kode (kode.ajith@gmail.com)

TURNING SUSTAINABLE AGTECH INTO PROFITABLE REALITY

Key Assumptions Behind Financial Model

****All price assumptions are based on market pricing as of June 2026****

CORE FINANCIAL & YIELD ASSUMPTIONS:		CAPITAL INVESTMENT ASSUMPTIONS:			
PARAMETER	ASSUMPTION	AREA	PARAMETER	ASSUMPTION	COST
LAND & BUILDING	Assumed available, difficult to factor in cost given the variation across locations	FACILITY INFRASTRUCTURE	Clean Room Wall Panel	50mm wash-safe PUF insulated panels for ceiling and walls (300 sq ft room shell footprint)	₹ 4,50,000
FACILITY FOOTPRINT	300 sq ft total, split into 2 facilities of 150 sq ft each for flowering & multiplication cycles		Anti-Microbial Floor Coatings	Heavy-duty epoxy floor sealing to manage high relative humidity	₹ 60,000
WHOLESALE PRICES	Grade 1 Saffron: Rs. 3,50,000 / kg Yellow Stamens (B2B Pollen): Rs. 500 / kg Purple Petals (B2B Cosmetic): Rs. 150 / kg Corms Resale (B2B Corms): Rs. 1,800 / kg		Subtotal [1]	Total Facility Investment	₹ 5,10,000
INITIAL SOWING STOCK	1,000 Premium Jumbo Corms (Size 11+ cm, 22 kg total)	EQUIPMENT	Inverter HVAC Units	Two independent units (One per room) equipped with low-ambient cooling controls for 8°C triggers	₹ 1,40,000
INITIAL SOWING COST	Sourced at wholesale for ₹2,200 per kg (Total asset value: ₹48,400)		Grow Racks	Custom 5-tier frames. Includes dry-tray shelves for Room A and slanted drainage tracks for Room B	₹ 1,20,000
FLOWER YIELD	3 flowers per corm in Year 1		LED Lighting Array	IP65 waterproof bars calibrated to support both rooms at full capacity	₹ 2,40,000
SPICE EXTRACTION YIELD	1,000 fresh purple flowers yield approximately 7 grams of dried Grade 1 saffron stigmas		Exhaust & Rack Fans	Includes 10 multi-speed clip-on rack fans to break microclimate boundary layers and 2 room-level inline duct exhaust fan systems	₹ 40,000
BIOMASS (PETALS) YIELD	1,000 flowers yield ~0.03 kg of dried petals		Automation Systems	Two 25 kg/hr ultrasonic humidifiers, digital solenoid sprinkler manifolds, and pure RO filter lines	₹ 1,20,000
STAMEN (POLLEN POD) YIELD	1,000 flowers yield ~0.003 kg of dried stamens		Subtotal [2]	Total Equipment Investment	₹ 6,60,000
CORM MULTIPLICATION	Factored at a conservative 2.8× multiplication rate per seasonal vegetative cycle	CORM INVENTORY	Jumbo Corms	22 kg of 11+ cm circumference corms	₹ 48,400
			Subtotal [3]	Initial Inventory	₹ 48,400
		OVERALL	Grand Total [1]+[2]+[3]	TOTAL INITIAL CAPITAL INVESTMENT	₹ 12,18,400

5 Year Financial Model (Based on assumptions in prior slide)

****All price assumptions are based on market pricing as of June 2026****

PROJECTED YoY YIELD					
YIELD	Year 1 (The Pilot)	Year 2 (Growth)	Year 3 (Break-Even)	Year 4 (Max Output)	Year 5 (Steady State)
Active Corms Planted	1,000	2,800	7,840	20,625	20,625
Total Flower Yield	3,000	8,400	23,520	61,875	61,875
Dry Saffron Yield (Grams)	21.0	58.8	164.6	433.1	433.1
Excess Corms Sold	0	0	1,327	37,125	37,125

PROJECTED YoY FINANCIALS:					
Performance Framework	Year 1 (The Pilot)	Year 2 (Growth)	Year 3 (Break-Even)	Year 4 (Max Output)	Year 5 (Steady State)
Corm Sale Revenue	₹ 0	₹ 0	₹ 35,829	₹ 10,02,375	₹ 10,02,375
Spice & Biomass Revenue	₹ 7,368	₹ 20,630	₹ 57,765	₹ 1,52,002	₹ 1,52,002
Total Revenue Earned	₹ 7,368	₹ 20,630	₹ 93,594	₹ 11,54,377	₹ 11,54,377
Total Facility OpEx	₹ 30,000	₹ 50,000	₹ 90,000	₹ 1,75,000	₹ 1,85,000
Net Operational Profit	- ₹22,632	- ₹29,370	₹3,594	₹ 9,79,377	₹ 9,69,377
Cumulative Wallet Balance	- ₹12,41,032	- ₹12,70,402	- ₹12,66,808	- ₹2,87,431	₹6,81,946

Key Resources

- **Primary Agronomic Research:** [Drying Kinetics and Essential Oil Profiles of Saffron Bio-Residues \(MDPI PMC\)](#) – Scientific validation for your 45°C petal drying parameters.
- **Automation Systems & Sensors:** [AC Infinity Environmental Controllers and Sensor Probes](#) – For your multi-point data tracking grids.
- **Organic Substrates & Buffers:** [Sikri Farms Certified Organic Liquid Calcium](#) – Sourcing for your 100L organic buffering matrix.